Bridging the Public-Private Discomfort Gap: A Data-Driven What-If Optimization Framework Implementation

April 13, 2026

1 Introduction

The proposed Reference and Decision Models, discussed in Secs. IV and V of the paper, have been implemented in a comprehensive software framework to validate the methodology on a real-world scenario. This section describes the implementation of the entire workflow and details the data processing pipeline. This framework integrates *OpenTripPlanner* (OTP) for the generation of trip alternatives, the calculation of the Generalized Discomfort GD, the Propensity $Prop$, the High-Impact Index $H - index$, and the evolutionary optimization, which covers the automatic candidates generation procedures for each intervention lever, and the genetic algorithm used to solve the global optimization problem.

— aggiungere descrizione delle successive sezioni —

2 Experimental Setup and Data Processing

The experimental analysis focuses on the metropolitan area of Cagliari, Italy. To enable the *What-If* simulation, the demand data with the supply-side data of the local PT operators and the OpenStreetMap road network graph, are integrated. These sources of data are described below in Sec. 2.1. The alternative itineraries are generated as described in the Sec. 2.2.

— aggiungere descrizione delle successive sotto-sezioni —

2.1 Datasets

In this work, two main datasets related to the metropolitan area of Cagliari (Italy) are considered: a) a dataset of trips recorded from citizens of Cagliari during the *ioPollicino* trial carried out in the period from April 2024 to June 2024 [1]; b) the GTFS data of the same period about offered services of two public transport agencies that operate in the Cagliari area (CTM [2] and ARST [3]). These datasets are illustrated in the following.

2.1.1 Observed Trips (Crowdsourced dataset from ioPollicino Trial)

The dataset used in this reasearch to describe the demand for mobility in Cagliari was recorded during the ioPollicino sustainable mobility trial [1]. ioPollicino is also the tracking system used during the trial on a sample of sharing mobility users in the cities of Cagliari and Bologna to collect and analyze citizens' travel habits. The system collects data on short proximity, concatenated, combined, interzonal trips, mode of transport and usage patterns. The trips are recorded semi-automatically, using a dedicated app, engaging the user with a gamification and rewards system. ioPollicino system consists of two parts: *Frontend*, the mobile tracking app and *Backend*. On first use, the app collects anonymous data about the user via a questionnaire. Then, the app records the user's movements in terms of GPS tracks (points, timestamps, and modes of transportation). Activity verification is proposed to the user (that can confirm or change for example the transport mode) by sending a notification to the smartphone. The collected data were organized in a dedicated database and released into a dataset that are used in this work. Data collected about users are anonymous and include age group, gender, city and neighbourhood of residence, and other. An amount of 1626 enrolled users of whom 1074 generated trips, allowing 69611 trips to be recorded over 67 days of the trial, from April 18, 2024, to June 23, 2024.

The dataset is structured by four csv files:

1. **users at the date [date]** that contains anonymous information about trial participants (anonymous user id, age, etc) until a given date;

2. **trips of the date [date]** that collects data about the trips made by users on a specific day of the trial (trip id, user id, transportation mode, starting coordinates, ending coordinates, trip duration, travelled distance, etc);

3. **users questionnaires at the date [date]** that contains answers to a questionnaire administered to users, collecting demographic information and personal preferences (user id, question, answer, targets);

4. **user targets** that contains the gamification experiment goals achieved by each user.

This dataset allowed to analyse the mobility behaviour of users and study the factors that could influence their choice of means of transport.

2.1.2 Open Data of Public Transportation Services

Information on the Public Transport offer can be downloaded from the Sardinia Region website on the open data page [4, 5]. The static GTFS data [6, 7] that can be downloaded from the timetables page of the open data are considered. This information is valid for several months and represents the services planned by each public transport company operating in a given area. In this first study, only information concerning two companies area considered (CTM SpA [2] and ARST SpA [3]). Furthermore, the data considered from all the companies services are limited to the geographical area of Cagliari, which is the area in which both Local Public Transport (LPT) companies operate. This dataset collects information on public transport companies and their services with the aim of mapping the public transport offer, in terms of transport routes, stops, and timetables. Its structure is organised in separate text files (.txt) and numerous information fields as it is described in the following [6]: 1) **agency** that contains id, name and url of the PT company providing the service; 2) **calendar** that defines the calendar of regular services; 3) $calendar_dates$ which specifies exceptions or variations to the regular calendar defined in the calendar file; 4) $fare_attributes$ which contains information on ride prices; 5) **routes** that contains information on the different transport routes and links to other dataset files; 6) **shapes** (optional) that provides information on the shape of routes, for displaying routes on a map; 7) **stops** provides details of the stops or stations in the service; 8) $stop_times$ describes the stop times for each trip; 9) **trips** represents the planned trips for each route; 10) $fare_rules$ (only for ARST) describes the rules for applying tariffs to specific lines or zones; 11) $feed_info$ (only for ARST) provides general information about the feed, such as the name. The structure offered by the GTFS data allows to map the entire PT system, to analyse route coverage and frequency, timetables at stops, variations on special days, and information on fares.

2.2 Estimation of travel itineraries

To evaluate the potential modal shift and compute the Generalized Discomfort metric defined in the Reference Models section (Sec. IV) of the paper, alternative itineraries (indexed as $k \geq 1$) for each observed trip ($k = 0$) must be generated. Specifically, PT alternatives for observed Private Vehicle (PV) trips, and vice versa, are computed.

This itinerary generation was performed using OpenTripPlanner (OTP) [8], an open-source multi-modal routing engine. OTP calculates optimal travel itineraries based on spatial-temporal constraints by combining street network graphs and transit schedules. To ensure computational efficiency and data consistency, a local OTP server configured with the OpenStreetMap (OSM) [9] network of Sardinia and the official GTFS static feeds from the local operators (CTM and ARST), released in the same period of the IoPollicino trial, has been deployed.

For each observed trip, the OTP routing API was queried to return up to a maximum of $N_{max} = 30$ alternative itineraries within the trip's departure time window. However, the raw output from OTP often includes redundant solutions. To ensure the quality and behavioral relevance of the choice set, we applied a strict post-processing filtering pipeline:

- **Duplicate structural filtering:** OTP frequently returns itineraries that share the exact same sequence of segments (same mode, length, and duration) but are merely shifted in time due to the transit schedule. We filtered out these time-shifted duplicates, retaining only unique structural itineraries, as redundant options do not add meaningful information to the discrete choice model.

- **Walking distance constraint:** To reflect realistic user behavior and physical constraints, we discarded any PT alternative where the total walking distance (access, transfer, and egress) exceeded 1,000 meters.

Following this generation and filtering process, the final dataset yielded approximately 32,000 valid PT alternative itineraries for the observed PV trips, and about 10,000 PV alternative itineraries for the observed PT trips. These clean itinerary sets serve as the foundational input for computing the generalized costs and evaluating the What-If scenarios.

2.3 Space-Time Discretization Parameters

Consistent with the model definition in Sec. IV of the main manuscript, we instantiated the discretization parameters to balance computational efficiency with spatial resolution:

- **Spatial Grid:** The urban territory is partitioned into square cells of size $c_{size} = 500$ meters. This is motivated by the fact that it is needed to consider cells of a size comparable to the maximum walking distance that was considered in the calculation of the alternative itineraries.
- **Time Slots:** The daily horizon is divided into slots $h = 10$ minutes. This resolution is sufficiently fine to model transfer synchronization but coarse enough to smooth out minor stochastic variations in departure times.

Consequently, the corridors $\mathcal{C}_{odh}$ are computed based on these specific quantization thresholds.

2.4 Generalized Discomfort Calculation

Once the valid travel itineraries are generated, the next implementation step involves the computation of the Generalized Discomfort metric for both the observed baseline ($k = 0$) and the simulated alternatives ($k \geq 1$). Rather than relying on simple geographical metrics, as Euclidean or Haversine distances which fail to capture the actual road network topology and multimodal constraints, our framework leverages the granular temporal segments extracted from the data sources.

Specifically, the data parsing pipeline is implemented to evaluate the following components:

- **Observed Trips ($k = 0$):** For the actual journeys recorded in the *IoPollicino* dataset, the total travel duration is derived directly from the start and end GPS timestamps. For private vehicle trips, access, egress, and waiting times are assumed to be negligible (zero), effectively representing a seamless door-to-door experience.
- **Alternative Itineraries ($k \geq 1$):** The JSON responses from the OTP API are systematically parsed to extract detailed segment durations. This includes the total walking time (aggregating the access to the first stop, transfers between lines, and egress to the final destination), the waiting times at transit stops, and the in-vehicle riding times.

These extracted temporal variables are then aggregated using the cost formulation defined in our theoretical model (see Sec. IV-B of the main manuscript. The physical effort and waiting penalties are properly weighted using the respective monetary Value of Time (β) coefficients.

To validate the implemented metric and highlight the existing competitiveness gap, Fig. 1 compares the distributions of the differential discomfort values computed for the two mobility profiles in the dataset. As expected, the ΔD values computed for actual PT trips (and their simulated PV alternatives) are structurally lower. Conversely, the distribution for observed PV trips compared against their PT alternatives exhibits a significantly higher median differential discomfort and is characterized by a long right tail. While this long right tail empirically highlights structural "transit deserts" - areas where PT is highly uncompetitive and massive infrastructural investments would be required - from an optimization perspective, the most crucial segment is the **central overlap region** between the two distributions. In this area, the discomfort gap between PV and PT is moderate, which corresponds exactly to the steepest section of the Sigmoid propensity curve defined in our theoretical model Sec IV-C of the main manuscript. These trips represent a *contestable market*: latent demand where even minor, cost-effective GTFS interventions (such as schedule synchronization or minor frequency adjustments) yield the highest marginal increase in mode-shift probability. Consequently, our What-If optimization solver naturally targets these high-sensitivity corridors to maximize the mode shift and the overall Return on Investment (ROI) of the network.

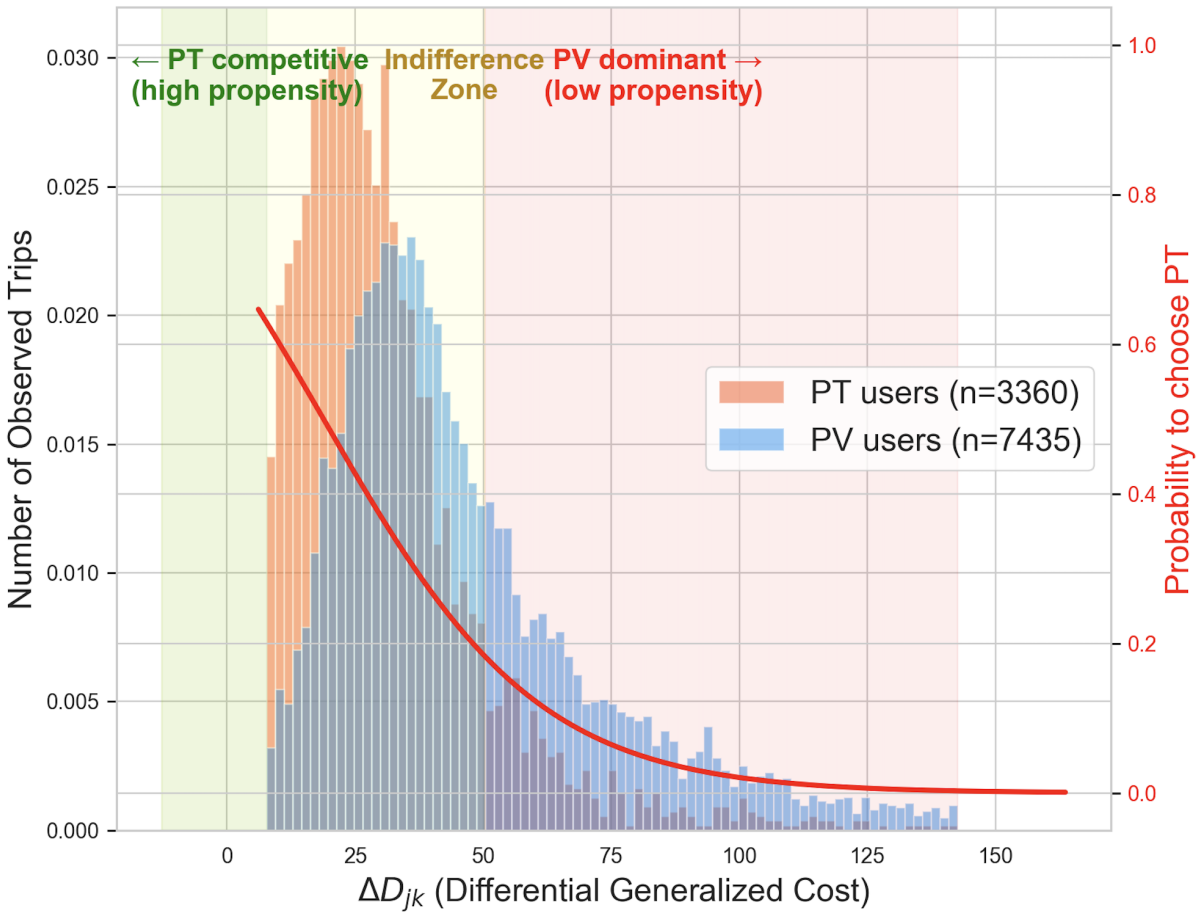

Figure 1: Distributions of the Differential Discomfort Metric values for observed PT trips vs PV itineraries (blue distribution) and observed PV trips vs PT itineraries (orange distribution).

2.5 Propensity Model and Sigmoid calibration

We give standard initial values for the parameters of the sigmoid function in order to use the $curve_fit$ function to fitting our data to the sigmoid model [10]. In particular, we do the following assumptions:

- the maximum propensity $P_{max} = Prop'_{max} = 1.0$, where we are assuming that theoretically all users could choose the bus. In nonlinear optimization algorithms such as Levenberg-Marquardt (used by $curve_fit$) [10] starting from an boundary of the allowable domain can help avoid local maximum when the target function has multiple solutions. If the true value of P_{max} is less than 1 (which might make sense in transport terms if users never all choose the bus), the algorithm will still converge towards the lowest corrected value. While if you start from a value that is too low, the algorithm could converge to a suboptimal local maximum.
- the gradient $g = g' = 1.0$ by which we assume an initial moderate slope. The gradient g of the sigmoid function controls the speed of the transition. A value of 1.0 represents a neutral scale with respect to the data.
- the flex point $\Delta D_0 = \Delta D'_0 = mean(x)$, where we assume that its initial value is taken as the mean value of D. Without a priori information on the distribution of the decision behavior, the sample mean $mean(x)$ is the best undistorted estimate of the "center" of the distribution of the observed ΔD_{jk} values.

Regarding the constraints for these parameters, we impose values suitable for our case study: 1) the P will be between 0 and 1, because the propensity cannot be less than 0 or greater than 100%; 2) the gradient will be between 0 and 10 to avoid unrealistic slopes (variability of propensity); 3) ΔD_0 will be included within the values of ΔD (i.e. $min() < \Delta D_0 < max()$). In this phase we train the sigmoid model on the calculated Discomfort data by associating it with the actual modal choice made by the user, estimating its parameters. In particular we obtain: $Prop_{max} = 1$ **(verificare questo valore)**, $g = 2.6198$ and $D_{flex} = 1.5160$. The trained model can be used for a variety of purposes, including: predicting the propensity values to change mode for new values of ΔD_{jk}, identifying critical thresholds where propensity becomes too low, and thus also to identify areas where it would be more convenient to intervene with changes to the PT services, but also to understand their impact in terms of modal shift.

2.6 H-index calculation

In order to simulate the improvement changes to PT only where they bring the greatest impact on the reduction of the Discomfort metric considered in the study, the H-index (High Impact Index) relative to each corridor $\mathcal{C}_{odh}$ considered was calculated. The goal is to identify where, but also when, public transport interventions bring the greatest benefits in terms of modal shift, then we need a way to combine two key factors: 1) low propensity (low likelihood of using PT services), and 2) high demand (i.e, high number of trips in a given area and time slot). It is possible to translate this goal in an objective function that maximises the modal shift potential (i.e. the number of users who could switch to public transport services). This potential, that we call *H-index* and we describe in the Paragraph c) is proportional to the total mobility demand (for a given geographical area and time period) and the potential for improvement, i.e. when the propensity can be increased. The latter is highest where the current propensity (calculated on actual data) is low but not zero, i.e. where the sigmoid function has the highest slope. In the next two paragraph we describe these two key factors.

Low Propensity By analyzing the sigmoid curve as a function of ΔD_{jk} we are able to identify the areas in which a given change in PT service can bring the greatest benefit in terms of increased choice of bus as a means of transport over private transport. That analysis has some important practical implications for our study, because it allows us to distinguish the area where the bus is very competitive with respect to the private vehicle, which is where $\Delta D_{jk} < 1.94$ (probability around 66%), instead for $\Delta D_{jk} > 3.30$ the public service is rarely chosen (probability is less than 3%), because it is too far from users' expectations. We focus on these trips that have ΔD_{jk} values between 1.94 and 3.30, which we consider *low propensity trips*.

High mobility demand In order to obtain a description of the total mobility demand space-time related Dem^h_{od}, we clusterize starting and ending points of the selected low propensity trips, obtaining a 3D Origin-Destination-Time matrix partitioning the space using a squared cell grid (500m x 500m) on the map of the city of Cagliari and the time by 10-minutes slots.

H-index This allows us to identify high-density-trip corridors (origin cell o, destination cell d, and time slot h), and calculate the *modal shift potential*, using a function "H-index" (1) defined as :

$$H^h_{od} = Dem^h_{od} * (1 - Prop^h_{od}) \tag{1}$$

where Dem^h_{od} is the mobility demand from cell A to cell B at time t and $Prop^h_{od}$ is the propensity to use public transport means in the same conditions. H-index is a suitable measure to identify where PT interventions bring the greatest benefits in terms of modal shift. We consider the product in the (1) represents the expected number of users who do not currently use public transport, but could potentially use it if the public service were improved.

Identification of priorities Sorting obtained corridors by decreasing H-index and selecting first N of them ($N = 5$ in our case), we obtain the *priority corridors* for a (simulated) change of the PT services and modified GTFS based on these corridors.

Simulation of improved routes - Distance Metric calculation we simulate improvements for the selected corridors by injecting ad-hoc modifications in the services, like assuming to add a new stop near to centroid of the cell A and/or cell B, or increasing the frequency for a given route. Then we calculate Distance Metric values in the new conditions.

Impact evaluation Using the sigmoid function, we calculate the new probabilities to choice the bus related to the new D_{ijk} values. Comparing these to the correspondant old D_{ijk} values, we calculate the *estimation of the (relative) modal shift*. We show initial results in the next section.

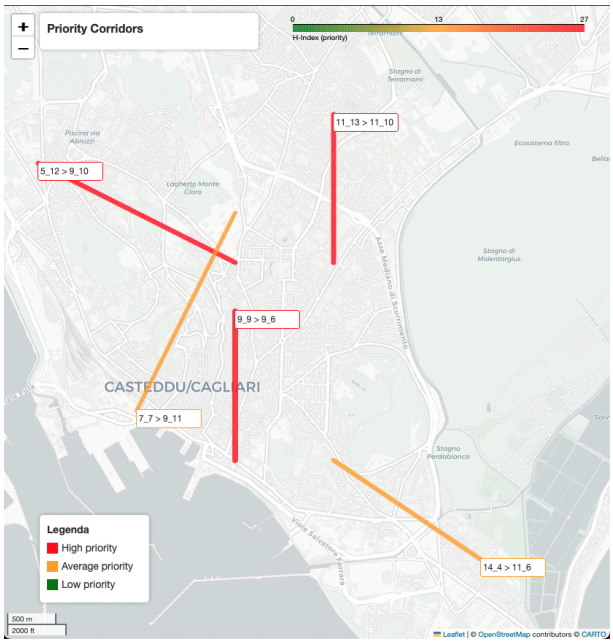

Figure 2: Priority corridors for public transport actions by multiplicative formula.

2.6.1 Optimization workflow

We divide the optimization workflow into two distinct phases:

- the automatic generation of feasible intervention candidates $\mathcal{Q}^\lambda$, based on transport engineering heuristics;
- the selection of the optimal subset of candidates via a Genetic Algorithm (GA), used to solve the global optimization problem formulated in Sec. V of the main manuscript. In addition, we provide the rationale for the choice of this particular algorithm.

Table 1: Results about simulation of actions on the transport system.

Corridors	**Main Issues**	*Dijk*	**Prop**	**Impact**
9-9 > 9-6 7:40 - 7:50 26.593	 Duration Walk dist	2.64 − > 1.90	0.05 − > 0.27	25.2%
11-13 > 11-10 8:30 - 8:40 20.858	 Duration Walk dist	2.62 − > 1.39	0.05 − > 0.58	54.89%
5-12 > 9-10 8:00 - 8:10 19.442	 Duration	2.86 − > 2.67	0.03 − > 0.05	2%
14-4 > 11-6 8:20 - 8:30 17.388	 Duration Walk dist	2.79 − > 1.97	0.03 − > 0.23	20%
7-7 > 9-11 22:50 - 23:00 14.055	 Frequency	2.55 − > 2.28	0.06 − > 0.12	6%

2.7 Algorithm Selection: Genetic Algorithm Rationale

The selection of the optimization method was driven by the specific mathematical properties of the decision model, which renders several traditional approaches unsuitable. The global optimization problem (Eq. 32 of the main manuscript) is non-linear, non-convex, and Mixed-Integer, then exhibits characteristics, that make traditional optimization methods unsuitable:

1. **Nature of Variables:** The decision vector $\mathbf{y}$ is purely binary, but the objective function involves complex interactions (e.g., the "sawtooth" shape of synchronization costs [11, 12] and quadratic speed costs [13]).

2. **High Dimensionality:** Decision vector $\mathbf{y}$ has $|\mathbf{y}| = |Q^{\texttt{stops}}| + |Q^{\texttt{freq}}| + |Q^{\texttt{speed}}| + |Q^{\texttt{sync}}| \approx$ several hundreds binary variables for medium-sized networks.

3. **Complexity:** the problem is combinatorial. As the number of candidates $|\mathcal{Q}|$ grow, the solution space grows exponentially $2^{|\mathcal{Q}|}$, making exact methods infeasible. With candidate-activation paradigm the space reaches $\prod_\lambda 2^{|Q^\lambda|}$ ($\approx 10^{60}$–10^{150} for realistic cases) requiring heuristic approaches.

4. **Non-Linearity:**
   - Frequency benefits exhibit diminishing returns: $\partial B^{\texttt{freq}}/\partial f_r \propto 1/f_r^2$
   - Speed costs are convex: $C_{ru} \propto (\Delta v_{ru})^2$

5. **Non-Convexity:** Schedule synchronization contains modulo operator (Eq. 24 of the main manuscript), creating sawtooth objective with local optima [14]

6. **Mixed Constraints:**
   - Linear budget constraints (Eq. 33 of the main manuscript)
   - Logical constraints (frequency ordering, offset selection)
   - Non-linear safety margins (Eq. 28 of the main manuscript)

Exact methods, such as Integer Linear Programming (ILP), were discarded primarily due to their inability to natively handle the non-convex objective function introduced by the synchronization modulo operator. Furthermore, the linearization required to adapt the problem to ILP solvers would inevitably degrade solution quality, while the computational time would grow exponentially with the problem size, making it infeasible for real-world instances. Similarly, gradient-based methods like Sequential Quadratic Programming are incompatible with the discrete, binary nature of the decision vector $\mathbf{y}$. Even if relaxed

to continuous variables, these methods are prone to entrapment in local optima within non-convex landscapes and lack the mechanism to handle the combinatorial activation logic inherent in the candidate selection. Simple greedy heuristics were also rejected as they fail to capture critical inter-lever synergies, such as the trade-off between stop rationalization and frequency increases, and cannot adequately manage the network locking effects required for effective schedule synchronization, leading to demonstrably suboptimal solutions.

In contrast, Genetic Algorithms (GAs) offer a robust meta-heuristic framework that is particularly well-suited for this specific optimization landscape [15]. Their natural affinity for binary encoding aligns perfectly with the proposed candidate-activation paradigm, allowing for the seamless handling of mixed variable types, including both binary topological decisions and integer synchronization offsets. By utilizing a population-based approach, the GA explores multiple regions of the solution space simultaneously, thereby mitigating the risk of convergence to local optima. Furthermore, GAs demonstrate superior flexibility in managing complex operational constraints through penalty functions and offer high scalability, as computational effort grows linearly with population size and benefits from inherent parallelizability. This suitability is further supported by the extensive and successful application of GAs in the literature regarding Transit Network Design Problems [15].

2.8 Candidate Generation Procedures

Before optimization, it is needed to generate the candidate sets $\mathcal{Q}^{\lambda}$, for each intervention lever through specialized preprocessing algorithms. These procedures leverage the spatial and temporal structure of the transportation network and the dataset of I_{jk} from Scenario Model (Sec. III of the main manuscript).

2.8.1 Stops Candidates ($\mathcal{Q}^{\texttt{stops}}$)

The generation of candidate stop locations requires balancing spatial coverage, commercial speed and operational feasibility. We adopt a methodology inspired by [16, 17, 18], extending it to support both stop addition and rationalization.

Input Data

- GTFS data: `routes.txt`, `trips.txt`, `shapes.txt` (route geometries), `stops.txt` (existing stops S^0)
- Road network graph $G = (V, E)$ from OpenStreetMap [9]
- Pedestrian network graph G^{walk} with sidewalk connectivity and crosswalks.
- Simulated alternative itineraries: origin/destination cells for priority corridors $\mathcal{C}_{priority}$

Algorithm **Step 1: Identify Active Route Segments.** For each priority corridor $\mathcal{C}_{odh} \in \mathcal{C}$, extract the set of routes $\mathcal{R}_{odh}$ composing alternative itineraries in the corridor. For each route $r \in \mathcal{R}_{odh}$:

1. Extract the geometric path (polyline) from GTFS `shapes.txt`
2. Identify "active arcs": segments of route r actually traversed by itineraries I_{jk} mapped to trips in $\mathcal{J}_{odh}$ (between o and d).
3. Construct $\mathcal{A}_{cand} = \bigcup_{r \in \mathcal{R}_{odh}} \mathcal{A}_r$, the union of all active arcs

Step 2: Discretize Active Arcs. For each arc $a \in \mathcal{A}_{cand}$ of length L_a:

1. Generate candidate points p at regular intervals δ_s **valori da controllare (e.g., 50-100 m)** along the route centerline
2. Project candidates onto nearest road network edge using spatial join
3. Snap to logical stop locations (intersections, existing infrastructure, commercial/residential areas)

This produces a "cloud" of potential stop locations p along bus routes serving priority corridors.

Step 3: Feasibility Filtering. A candidate point p is added to the final set S^{new}, becoming a candidate stop s_c if and only if it satisfies operational constraints:

1. *Minimum spacing from existing stops:*

$$\min_{s \in S^0} \{d(s_p, s)\} \geq \Delta d_{min} \tag{2}$$

   where: a) s is the location of the esisting stop; b) p is the new candidate point; c) $d(\cdot)$ is network distance; and Δd_{min} (e.g., 200-300 m) maintains commercial speed and reduces operational instability [16].

2. *Accessibility:* Candidate must be reachable via pedestrian network G^{walk} from surrounding cells

3. *Safety:* Exclude locations with insufficient visibility, high-speed roads without crossings, or physical barriers

Step 4: Pre-Compute Access Benefits (lookup table). For each generated candidate $s_c \in S^{new}$ and each pair $(j, 0)$ - (j, k) :

1. Compute walking distance $d_{jk}^{w,via-s_c}$ using Haversine formula or network routing;

2. Store projected access benefit:

$$\Delta d_{jk-s_c}^{w} = \max\left(0, d_{jk}^{w,0} - d_{jk}^{w,via-s_c}\right) \tag{3}$$

3. Pre-filter: Remove candidates where $\sum_{jk} \Delta d_{jk-s_c}^{w} <$ threshold (minimal aggregate benefit).

Step 5: Estimate Dwell Time Penalty. For each candidate s_c, estimate dwell time τ_{s_c} **controllare questi valori (typically 20-40 seconds)** [19], based on expected boarding/alighting demand from nearby cells.

Output

- Candidate set $Q^{\texttt{stops}} = S^0 \cup S^{new}$ **controllare questi valori with** $|Q^{stops}| \approx$ 50-200 **per corridor**;
- Lookup tables: $\Delta d_{jk-s_p}^{w}$ (access benefits), τ_{s_c} (dwell penalties)

Computational Complexity $O(|\mathcal{R}| \cdot L_r/\delta_s + |\mathcal{J}| \cdot |Q^{\texttt{stops}}|)$ where L_r is average route length.

2.8.2 Frequency Candidates ($\mathcal{Q}^{\texttt{freq}}$)

Frequency candidates represent discrete vehicle additions, following the principle that service improvements should be *incrementally deployable* [20].

Input Data

- GTFS `frequencies.txt` or `stop_times.txt`: Baseline frequency $f_r(\mathbf{0})$
- Fleet constraints: Maximum available vehicles K_{max}^{fleet}
- Operational parameters: Minimum safe headway H_{min} (e.g., 5 minutes), vehicle capacity, cycle time

Algorithm **Step 1: Extract Baseline Frequencies.** For each route $r \in \mathcal{R}_{odh}$:

- If `frequencies.txt` exists: read $f_r(\mathbf{0})$ directly
- Otherwise: infer from scheduled departures in `stop_times.txt`: $f_r(\mathbf{0}) \approx N_{trips}/T_{service}$

Step 2: Determine Maximum Feasible Frequency.

$$f_r^{max} = \min\left\{\frac{1}{H_{min}},\ 3 \times f_r(\mathbf{0})\right\} \tag{4}$$

The first term ensures safety (minimum headway), the second avoids unrealistic service levels **Questi valori sono ragionevoli, ma di esempio per il calcolo della complessità computazionale.**

È difficile inserire una reference per tutti questi valori, costanti e pesi, forse è meglio rimuoverli..

Step 3: Generate Discrete Increments. For route r, create $Q_r^{\texttt{freq}} = \{\Delta f_{r,1}, \Delta f_{r,2}, ..., \Delta f_{r,Q_r}\}$ where:

- $\Delta f_{r,q} = q$ (each candidate adds q vehicles)
- $Q_r = \lfloor f_r^{max} - f_r(\mathbf{0}) \rfloor$ (maximum addable vehicles)

Step 4: Pre-Compute Wait Time Reductions. For each routes r associated to the itinerary I_{jk}:

- Baseline wait: $t_{jk-r}^{wait}(\mathbf{0}) = 1/(2f_r)(\mathbf{0})$
- For each increment q: $t_{jk-r}^{wait}(\mathbf{y}_{rq}^{\texttt{freq}}) = 1/(2(f_r(\mathbf{y}^{\texttt{freq}}{}_{rq})))$
- Store benefit: $\Delta t_{jk-r}^{wait}(\mathbf{y}_{rq}^{\texttt{freq}}) = t_{jk-r}^{wait,0} - t_{jk-r}^{wait}(\mathbf{y}_{rq}^{\texttt{freq}})$

Output

- $Q^{\texttt{freq}} = \bigcup_r Q_r^{\texttt{freq}}$
- Lookup table: $\Delta t_{jk-r}^{wait}(rq)$ for all (j, k, r, q) combinations

2.8.3 Speed Optimization - Infrastructure Intervention Catalog

Speed improvement candidates are *predefined infrastructure upgrades* with known costs and expected speed gains, drawing from engineering catalogs and Bus Rapid Transit (BRT) literature [13, 21].

Intervention Types We define three standard intervention categories:

Type 1: Transit Signal Priority (TSP)

- Applicability: Signalized intersections on arterial roads
- Speed gain: $\Delta v_{ru} \approx$ 2-4 km/h [21]
- Cost: $C_{ru} = N_{signals} \times 50000$ euro (ITS equipment per intersection)

Type 2: Dedicated Bus Lanes

- Applicability: Arterial roads with ≥ 4 lanes
- Speed gain: $\Delta v_{ru} \approx$ 5-10 km/h [13]
- Cost: $C_{ru} = L_r \times 200.000$ euro/km (striping, signage, enforcement cameras)

Type 3: Queue Jump Lanes

- Applicability: High-delay intersections with space for short bus-only lane
- Speed gain: $\Delta v_{ru} \approx$ 3-6 km/h
- Cost: $C_{ru} = N_{junctions} \times 100.000$ euro (lane reconfiguration, signals)

Algorithm **Step 1: Segment Routes into Intervention Zones.** For each route $r \in \mathcal{R}_{odh}$, divide path into segments based on:

- Road functional class (from OpenStreetMap)
- Congestion level (from GPS speed data or traffic models)
- Existing infrastructure (mixed traffic vs. dedicated)

Step 2: Map Interventions to Segments. For each segment to upgrade u of route r:

1. Identify applicable intervention types (based on segment characteristics)
2. Estimate speed gain Δv_{ru} using local calibration or literature values

3. Compute cost C_{ru} using local infrastructure pricing and segment length

Step 3: Pre-Compute Fleet Savings. For each route r and intervention segment u:

- New cycle time: $T_r^{cycle}(u) = L_r/(v_r(\mathbf{0}) + \Delta v_r(\mathbf{y}_{ru}^{\texttt{speed}})) + t_{layover}$
- New fleet requirement: $N_r(u) = \lceil f_r(u) \cdot T_r^{cycle}(u) \rceil$
- Fleet savings: $\Delta N_r(u) = N_r(\mathbf{0}) - N_r(\mathbf{y}_{ru}^{\texttt{speed}})$

Output

- $Q^{\texttt{speed}} = \bigcup_r \mathcal{Q}_r^{\texttt{speed}}$
- Parameter tables: Δv_{ru}, C_{ru}, $\Delta N_r(u)$

2.8.4 Schedule Synchronization - Temporal Discretization

Synchronization candidates are *temporal offsets* applied to route schedules to minimize transfer waiting times [14, 12].

Input Data

- the set of transfer legs $\mathcal{T}_{odh}$ for each priority corridor (from OTP simulation)
- Route headways H_r (from GTFS data)
- Current schedules: τ_{jkl}^s, τ_{jkl}^e, t_{jkl}^{wk} for all $\ell_{jkl} \in \mathcal{T}$ (being $\mathcal{T}$ the set of all `tr` type legs).
- Parameters: Δ (offset step, e.g. 1-2 minutes) [14], t^{safe} (safety margin, e.g 2-3 minutes) [14].

Algorithm **Step 1: Identify Critical Transfers.** For each transfer leg $\ell_{jkl} \in \mathcal{T}$:

1. Compute baseline slack: $t_{jkl}^{wt}(\mathbf{0}) = \tau_{jkl}^e - \tau_{jkl}^s - t_{jkl}^{wk}$
2. Flag as critical if: $t_{jkl}^{wt}(\mathbf{0}) > 8$ minutes
3. Assign importance weight: $w_{jkl}^{trans} = w_{jk}$,

Step 2: Extract Involved Routes.

$$\mathcal{R}_T = \{r \in \mathcal{R} : \exists\, \ell_{jkl} \in \mathcal{T} \text{ s.t. } r \in \{r_{arr}(\ell_{jkl}), r_{dep}(\ell_{jkl})\}\} \tag{5}$$

Step 3: Discretize Offset Ranges. For each route $r \in \mathcal{R}_T$ with headway H_r:

1. Number of offsets: $K_r = \lfloor H_r/\Delta \rfloor$
2. Generate candidates: $\delta_{r,\theta} = -H_r/2 + \theta \cdot \Delta$ for $\theta = 0, 1, ..., K_r$
3. Ensure $\delta_{r,\theta^*} = 0$ is included (baseline, no shift)
4. Build $Q_r^{\texttt{sync}} = \{\delta_{r,0}, \delta_{r,1}, ..., \delta_{r,K_r}\}$

Step 4: Pre-Compute Transfer Wait Times (Lookup Table). For each transfer leg $\ell_{jkl} \in \mathcal{T}$ and offset pair (σ, ξ) where $\sigma \in Q_{r_{arr}(\ell_{jkl})}^{\texttt{sync}}$, $\xi \in Q_{r_{dep}(\ell_{jkl})}^{\texttt{sync}}$:

$$T_{jkl,\sigma,\xi}^{wait} = \left[(\tau_{jkl}^e + \delta_{r_{dep},\xi}) - (\tau_{jkl}^s + \delta_{r_{arr},\sigma}) - t_{jkl}^{wk}\right] \bmod H_{r_{dep}} \tag{6}$$

This $O(|\mathcal{T}| \times K_{max}^2)$ table enables $O(1)$ fitness evaluation during GA iterations.

Step 5: Identify Safety Violations. Mark offset combinations where $T_{jkl,\sigma,\xi}^{wait} < t^{safe}$ for constraint handling in GA.

Output

- $Q^{\texttt{sync}} = \bigcup_{r \in \mathcal{R}_T} Q_r^{\texttt{sync}}$
- lookup table: $T_{\ell_{jkl},\sigma,\xi}^{wait}$
- Safety violation flags for fitness penalty

Computational Optimization For large networks ($|\mathcal{T}| > 5000$), use sparse storage (only store non-zero transfer times) and batch compute in parallel. **Solo il 34 per cento delle alternative per Cagliari presenta trasbordi. Verificare il numero esatto di trasbordi nel dataset delle trip-alternatives, per avere una indicazione di un case study (Cagliari).**

Therefore, we adopt a GA approach tailored to the binary activation vector.

2.9 Genetic Algorithm Description

This section details the Genetic Algorithm (GA) designed to solve the global optimization problem formulated in Eq. 32 of the main manuscript. . GAs have been extensively applied to transit network design problems due to their ability to handle combinatorial, non-convex solution spaces [22, 23]. The algorithm is structured around the candidate-activation paradigm, where the chromosome directly encodes the binary activation status of all intervention candidates, tailored to handle the binary decision vector $\mathbf{y}$ and then multi-lever intervention framework.

2.9.1 Chromosome Encoding

Each individual chromosome is represented as a concatenated binary vector encoding the activation status of all candidates across the four intervention levers:

$$\mathbf{y} = \left[\mathbf{y}^{\texttt{stops}} \mid \mathbf{y}^{\texttt{freq}} \mid \mathbf{y}^{\texttt{speed}} \mid \mathbf{y}^{\texttt{sync}}\right] \in \{0,1\}^{|Q|} \tag{7}$$

where $|Q| = |Q^{\texttt{stops}}| + |Q^{\texttt{freq}}| + |Q^{\texttt{speed}}| + |Q^{\texttt{sync}}|$ is the total number of candidate interventions across all levers. The binary structure naturally fits the candidate-activation paradigm: $y_q = 1$ indicates that candidate intervention q is activated in the solution. This concatenated encoding ensures that crossover and mutation operators can simultaneously explore different intervention combinations.

2.9.2 Population Initialization

The initial population $\mathcal{P}^{(0)}$ of size N_{pop} is generated using a hybrid strategy to balance exploration and domain knowledge [23]:

1. **Random Generation:** The majority of the population is initialized uniformly at random. Each bit y_q is assigned a 0 or 1 with equal probability, ensuring a broad exploration of the solution space.
2. **Heuristic Seeding:** The remaining portion is seeded with domain-specific solutions to accelerate convergence. This includes the baseline scenario (all zeros, $\mathbf{y} = \mathbf{0}$) and greedy solutions that activate only the least expensive interventions on the corridors with the highest H^0_{odh} index.

2.9.3 Fitness Evaluation and Constraint Handling

The unconstrained objective of the GA is to maximize the Net Utility $Z_{PT}(\mathbf{y})$. However, the presence of hard operational and budgetary constraints (e.g., Eq. 33 of the main manuscript) requires a specific handling mechanism. While more sophisticated, parameter-free constraint handling techniques exist [24], we adopt a *static penalty* approach. This method remains widely used in Genetic Algorithms due to its simplicity, ease of implementation, and recognized robustness. This approach transforms the constrained problem into an unconstrained one by modifying the fitness function F(y):

$$F(\mathbf{y}) = Z_{PT}(\mathbf{y}) - \sum_{c \in \mathbb{C}} \omega_c \cdot \max(0, g_c(\mathbf{y})) \tag{8}$$

where $\mathbb{C}$ is the set of constraints, $g_c(\mathbf{y})$ measures the magnitude of the violation for constraint c (e.g., the budget excess in euros), and ω_c is a severe penalty weight. If an individual violates a constraint, its fitness is drastically reduced, effectively excluding it from the mating pool while allowing the algorithm to safely explore the boundaries of the feasible region.

2.9.4 Genetic Operators and Execution Parameters

To generate the offspring population $\mathcal{P}^{(t+1)}$ from the current generation $\mathcal{P}^{(t)}$, the following standard operators are applied [25]:

- **Selection:** A Binary Tournament Selection (size $k = 3$) is used. This method provides adequate selection pressure toward fitter individuals without causing premature convergence.
- **Crossover:** A Uniform Crossover operator is applied with a probability $p_c = 0.8$. Given the concatenated nature of the chromosome (Eq. 7), uniform crossover allows the independent mixing of intervention levers (e.g., inheriting speed interventions from parent A and frequency from parent B).
- **Mutation:** A Bit-Flip Mutation is applied with a probability $p_m = 1/|Q|$ per gene, where $|Q|$ is the chromosome length. As theoretically demonstrated [26], this specific mutation rate is optimal for maximizing the probability of improving a nearly optimal string without disrupting already correct bits, effectively flipping on average one candidate activation per chromosome.

2.9.5 Stopping Criteria

The evolution process terminates when either of the following criteria is met: 1) a maximum number of generations N_{gen} (e.g., 200) is reached, or 2) the fitness of the best individual in the population does not improve by more than a tolerance threshold ϵ for 30 consecutive generations, indicating that the algorithm has likely converged to a stable, near-optimal solution.

References

[1] Cagliari Digital Lab, "Iopollicino," January 2025. [Online]. Available: https://www.cagliaridlab.it/it/iopollicino.page

[2] "Ctm spa, cagliari," January 2025. [Online]. Available: https://www.ctmcagliari.it

[3] "Arst spa," January 2025. [Online]. Available: https://www.arst.sardegna.it

[4] Regione Sardegna, "Open data, regione sardegna," December 2024. [Online]. Available: https://www.sardegnamobilita.it/open-data

[5] ——, "Services, regione sardegna," December 2024. [Online]. Available: https://www.sardegnamobilita.it/open-data/offerta-servizi

[6] "Gtfs reference," March 2026. [Online]. Available: https://gtfs.org/documentation/schedule/reference/

[7] CTM SpA, "Ctm gtfs page," January 2025. [Online]. Available: https://www.ctmcagliari.it/opendata/

[8] "Opentripplanner," January 2025. [Online]. Available: https://www.opentripplanner.org

[9] "Openstreetmap.org," January 2025. [Online]. Available: https://www.openstreetmap.org/about

[10] P. e. a. Virtanen, "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.

[11] A. Ceder, *Public Transit Planning and Operation: Theory, Modeling and Practice.* Butterworth-Heinemann/Elsevier, 2007.

[12] M. H. Rapp and C. D. Gehner, "Transfer optimization in an interactive graphic system for transit planning," W. and J. Rapp Company / University of Washington, Tech. Rep., 1976.

[13] G. Currie, "Bus transit oriented development - strengths and challenges relative to rail," *Journal of Public Transportation*, vol. 9, no. 4, pp. 1–21, 2006.

[14] A. Ceder, *Public Transit Planning and Operation: Modeling, Practice and Behavior.* CRC Press, 2016.

[15] W. Fan and R. Machemehl, "Optimal transit route network design problem with variable transit demand: genetic algorithm approach," *Journal of Transportation Engineering*, 2006.

[16] P. G. Furth and A. B. Rahbee, "Optimal bus stop spacing through dynamic programming and geographic modeling," *Transportation Research Record*, vol. 1731, no. 1, pp. 15–22, 2000.

[17] A. Ibeas, L. dell'Olio, B. Alonso, and O. Sainz, "Optimizing bus stop spacing in urban areas," *Transportation Research Part E: Logistics and Transportation Review*, vol. 46, no. 3, pp. 446–458, 2010.

[18] A. T. Murray, "Strategic analysis of public transport coverage," *Socio-Economic Planning Sciences*, vol. 35, no. 3, pp. 175–188, 2001.

[19] A. Tirachini, "The economics and engineering of bus stops: Spacing, design and congestion," *Transportation Research Part A*, vol. 59, pp. 37–57, 2014.

[20] E. E. Osuna and G. F. Newell, "Control strategies for an idealized public transportation system," *Transportation Science*, vol. 6, no. 1, pp. 52–72, 1972.

[21] H. S. Levinson, "Analyzing transit travel time performance," *Transportation Research Record*, vol. 915, pp. 1–6, 1983.

[22] P. Chakroborty, "Genetic algorithms for optimal urban transit network design," *Computer-Aided Civil and Infrastructure Engineering*, vol. 18, no. 3, pp. 184–200, 2003. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1111/1467-8667.00309

[23] V. Guihaire and J.-K. Hao, "Transit network design and scheduling: A global review," *Transportation Research Part A: Policy and Practice*, vol. 42, no. 10, pp. 1251–1273, 2008. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0965856408000888

[24] K. Deb, "An efficient constraint handling method for genetic algorithms," *Computer Methods in Applied Mechanics and Engineering*, vol. 186, no. 2–4, pp. 311–338, 2000.

[25] A. E. Eiben and J. E. Smith, *Introduction to Evolutionary Computing*, 2nd ed. Berlin, Heidelberg: Springer, 2015, standard textbook on evolutionary algorithms; covers selection, crossover, and mutation operators.

[26] H. Mühlenbein, "How genetic algorithms really work: Mutation and hillclimbing," in *Parallel Problem Solving from Nature — PPSN II*, R. Männer and B. Manderick, Eds. Amsterdam: Elsevier, 1992, pp. 15–25.